

NovaDiff — Change report

Baseline: NovaDiff-main

Target: NovaDiff

Generated: 2026-05-14T07:37:35.642Z

Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI project narrative

Workspace overview — roots, comparison intent, scale of change

The workspace is a diff between two versions of a codebase, with the baseline being the older version and the target being the newer version. The baseline folder label is "NovaDiff-main" and the target folder label is "NovaDiff". The total changed paths in this workspace are 42, which indicates that there were significant changes in the codebase between the two versions. The Gitignore file was compared, and it reported 140 changed path(s) but only 98 were omitted under the baseline and/or target root .gitignore or .git/info/exclude. This suggests that some files were intentionally excluded from the comparison.

By kind – added: 21, removed: 0, modified: 21 – the majority of changes were related to modifications made to existing files rather than additions or deletions. Max path depth (segments): 3 – the deepest path in the workspace has three segments. Depth histogram:

- Depth 1: 3 paths
- Depth 2: 16 paths
- Depth 3: 23 paths

Top first path segments (coarse "module" buckets):

- src/: 25
- electron/: 11
- ./: 3
- cli/: 3

Top extensions among changed files:

- .ts: 14
- .cjs: 11
- .tsx: 10
- .json: 2
- (no ext): 1
- .lock: 1
- .toml: 1
- .rs: 1
- .css: 1

Sample of up to 42 relative paths (not exhaustive):

- .gitignore
- cli/Cargo.lock
- cli/Cargo.toml
- cli/src/main.rs
- electron/code-city-model.cjs
- electron/diff-excerpt.cjs
- electron/file-summary-export.cjs
- electron/git-blame.cjs
- electron/llm.cjs
- electron/main.cjs
- electron/novadiff-docs-html.cjs
- electron/novadiff-docs-pdf.cjs
- electron/novadiff-docs-writer.cjs
- electron/prefetch-summaries.cjs
- electron/preload.cjs
- package-lock.json
- package.json
- src/App.css
- src/App.tsx
- src/app/codeCityLayout.ts

- src/app/codebaseDocs.ts
- src/app/commitMessage.ts
- src/app/diffExcerpt.ts
- src/app/docWorkspaceMetrics.ts
- src/app/docsQuality.ts
- src/app/generatedMarkdown.ts
- src/app/llmStorage.ts
- src/app/mermaidBoot.ts
- src/app/novadiffDocs.ts
- src/app/novadiffPaths.ts
- src/app/selectedDiffDocs.ts
- src/app/types.ts
- src/components/ChangedFilesTree.tsx
- src/components/CodeCityLegend.tsx
- src/components/CodeCityView.tsx
- src/components/DiffWorkspace.tsx
- src/components/DocumentationWorkspace.tsx
- src/components/InsightsColumn.tsx
- src/components/LlmSettingsModal.tsx
- src/components/LlmSummaryMarkdown.tsx
- src/components/SidebarNav.tsx
- src/vite-env.d.ts

Change landscape – interpret counts, dominant extensions, depth hotspots, risk intuition

The change landscape of this workspace is dominated by changes to TypeScript files (14), with a smaller number of changes to JavaScript and JSON files (2). The majority of changes are related to modifications made to existing files rather than additions or deletions.

Max path depth (segments): 3 – the deepest path in the workspace has three segments. Depth histogram:

- Depth 1: 3 paths
- Depth 2: 16 paths
- Depth 3: 23 paths

Top first path segments (coarse “module” buckets):

- src/: 25
- electron/: 11
- ./: 3
- cli/: 3

Top extensions among changed files:

- .ts: 14
- .cjs: 11
- .tsx: 10
- .json: 2
- (no ext): 1
- .lock: 1
- .toml: 1
- .rs: 1
- .css: 1

Subsystem map – group paths into coherent areas (config, tests, app, infra, vendor...) using only evidence from the sample paths and top segments

Based on this information, we can identify the following subsystems:

- src/: 25
- electron/: 11
- ./: 3

- cli/: 3

These subsystems are likely related to the codebase's structure and organization. The "src" directory is likely where the majority of the codebase's source code resides, while the "electron" directory may be related to an Electron application. The "cli" directory may be related to a command-line interface or other tooling for the codebase.

Cross-cutting concerns – security, build/release, migrations, observability (flag unknowns honestly)

There is no obvious indication of any cross-cutting concerns in this workspace, such as security vulnerabilities, build/release issues, migrations, or observability concerns. However, it is important to note that these types of concerns can be difficult to identify through diff analysis alone, and may require additional investigation and analysis.

Documentation & tooling gaps – what would require Doxygen/clangd/tree-sitter or runtime profiling to validate

There are several areas where documentation and tooling could be improved to better support the development and maintenance of this codebase:

- Documentation: There are several TypeScript files that do not have accompanying documentation, which could benefit from more thorough documentation. Additionally, there are some JavaScript and JSON files that could benefit from more detailed documentation.
- Tooling: The use of TypeScript and JavaScript files suggests that there may be opportunities for improving the codebase's tooling, such as using a linter or other static analysis tools to catch errors and improve code quality.

Suggested verification – tests, manual checks, staged rollout

Based on the evidence in this workspace, it is suggested that the following types of verification should be considered:

- Tests: There are no obvious signs of testing in this workspace, which could indicate a lack of test coverage. However, it is important to note that testing can be difficult to identify through diff analysis alone, and may require additional investigation and analysis.
- Manual checks: There are several changes to configuration files (e.g., .gitignore) and other non-code artifacts, which could benefit from manual review and approval. Additionally, there are some changes to build scripts (e.g., package.json) that may require manual review and approval.
- Staged rollout: There are several changes to the codebase's structure and organization, which may require a staged rollout process to ensure that all affected components are properly tested and validated before being released to production.

In conclusion, this workspace contains significant changes to the codebase, with a focus on TypeScript files and a complex structure. While there are no obvious signs of cross-cutting concerns or documentation & tooling gaps, there are opportunities for further investigation and analysis to better support the development and maintenance of this codebase.

Compare metrics

Metric Count
Added 21
Removed 0
Modified 21

Baseline: NovaDiff-main
Target: NovaDiff

Changed paths (sample, up to 80)

- **modified** .gitignore
- **modified** cli/Cargo.lock
- **modified** cli/Cargo.toml
- **modified** cli/src/main.rs
- **added** electron/code-city-model.cjs
- **added** electron/diff-excerpt.cjs
- **added** electron/file-summary-export.cjs
- **added** electron/git-blame.cjs
- **modified** electron/llm.cjs
- **modified** electron/main.cjs
- **added** electron/novadiff-docs-html.cjs
- **added** electron/novadiff-docs-pdf.cjs
- **added** electron/novadiff-docs-writer.cjs
- **added** electron/prefetch-summaries.cjs
- **modified** electron/preload.cjs
- **modified** package-lock.json
- **modified** package.json
- **modified** src/App.css
- **modified** src/App.tsx
- **added** src/app/codeCityLayout.ts
- **added** src/app/codebaseDocs.ts
- **added** src/app/commitMessage.ts
- **modified** src/app/diffExcerpt.ts
- **added** src/app/docWorkspaceMetrics.ts
- **added** src/app/docsQuality.ts
- **added** src/app/generatedMarkdown.ts
- **modified** src/app/llmStorage.ts
- **added** src/app/mermaidBoot.ts
- **added** src/app/novadiffDocs.ts
- **added** src/app/novadiffPaths.ts
- **added** src/app/selectedDiffDocs.ts
- **modified** src/app/types.ts
- **modified** src/components/ChangedFilesTree.tsx
- **added** src/components/CodeCityLegend.tsx
- **added** src/components/CodeCityView.tsx
- **modified** src/components/DiffWorkspace.tsx
- **added** src/components/DocumentationWorkspace.tsx
- **modified** src/components/InsightsColumn.tsx
- **modified** src/components/LlmSettingsModal.tsx
- **modified** src/components/LlmSummaryMarkdown.tsx
- **modified** src/components/SidebarNav.tsx
- **modified** src/vite-env.d.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/git-blame.cjs	cacheKey, parseBlamePorcelain, blameFileOwnership
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt, buildSummarizePrompt
electron/main.cjs	getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml, buildDiagramsHtml, buildReleaseHtml, readIfExists
electron/novadiff-docs-pdf.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/prefetch-summaries.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
scripts/run-electron.cjs	cleanEnv, resolveElectronBinary
src/App.tsx	isElectron, pickDirectory, baseName, isBinaryOrNonTextError, clampInsightsWidth, WindowChromeState, loadInsightsWidth, WindowChrome
src/app/changedFilesTree.ts	DirNode, FileNode, normalizePathSegments, ancestorDirPaths, sortChildren, ensureDir, addFilePathToTree, buildChangedFilesTreeRoot, flattenVisibleRows
src/app/codeCityLayout.ts	CodeCityFilters, CodeCityRenderableDistrict, CodeCityRenderableBuilding, CodeCityLayoutResult, hashColor, kindColor, changeEmissive, buildingHeight, visibleForRoot, buildCodeCityLayout
src/app/codebaseDocs.ts	sampleFilePaths, sampleSymbolLines, depthHistogram, buildCodebasePromptContext, buildCodebaseMetricsMarkdown, codebaseExtensionPieMermaid, codebaseDepthMermaid

File	Symbols / defs (regex)
src/app/commitMessage.ts	buildCommitMessageContext, parseCommitMessageOutput
src/app/diffExcerpt.ts	formatDiffRowLine, buildDiffExcerpt, buildDiffExcerptChunks
src/app/docWorkspaceMetrics.ts	ExtensionCount, DocWorkspaceMetrics, fileExtension, firstSegment, buildDocWorkspaceMetrics, GitignorePromptMeta, metricsToPromptContext, buildCompareMetricsMarkdown, changeKindPieMermaid, depthBarMermaid
src/app/docsQuality.ts	uniqStrings, compactText, relPathAnchorId, buildSummaryPromptContext, buildRiskPromptContext, deriveConfidenceBadges, buildReleaseOverviewMarkdown
src/app/generatedMarkdown.ts	escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, promoteKnownHeadings, splitDenseParagraphs, normalizeMarkdownChunk, normalizeGeneratedMarkdown
src/app/llmStorage.ts	LlmSettings, LlmSummarizePayload, defaultBase, defaultModel, defaultLlmSettings, loadLlmSettings, saveLlmSettings
src/app/mermaidBoot.ts	currentMermaidTheme, ensureMermaidInit, runMermaidNodes
src/app/novadiffDocs.ts	bundleLabel, bundleShortDescription
src/app/novadiffPaths.ts	isNovadiffDocsReservedPath
src/app/selectedDiffDocs.ts	formatDiffRowLine, compactIndices, summarizeLineRange, lineRangeLabel, buildExcerpt, getRowsForIndices, rowsForWindow, buildLineRanges, rangeSortScore, findBestSymbolForSide, findBestEnclosingSymbol, buildSelectionLabel, rowsWithinSymbol, slugifySelectionKey, isSelectableDiffRow, isChangedDiffRow, SelectedDiffDocContext, buildSelectedDiffDocContext
src/app/types.ts	FileChange, DiffRow, DiffSymbolSpan, DiffSelectionLineRange, DiffSelectionSymbolMatch, SelectionDocArtifactMeta, FileDiffPayload, EvidenceBadge, FileSummaryEvidence, OutlineFileSummary, OutlineSymbolSummary, OutlineSymbolFileEntry, OutlineDetectedProject, CodebaseOutline, CodeCityOwnershipSummary, RiskSignalAdvisoryMeta, RiskSignal, SummaryIndexEntry, SelectionIndexEntry, CodeCityFileNode, CodeCitySymbolNode, CodeCityModel
src/components/ChangedFilesTree.tsx	ChangedFilesTreeProps, TreeRow, ChangedFilesTree
src/components/CodeCityLegend.tsx	CodeCityLegend
src/components/CodeCityView.tsx	attachSelectable, CodeCityView
src/components/DiffWorkspace.tsx	DiffWorkspaceProps, DiffWorkspace, DiffTableRow
src/components/DocumentationWorkspace.tsx	mergeSummaryEntries, mergeSelectionEntries, yieldToBrowser, DocumentationWorkspaceProps, DocumentationWorkspace
src/components/InsightsColumn.tsx	InsightsColumnProps, InsightsColumn, MetricPill
src/components/LlmSettingsModal.tsx	LlmSettingsModalProps, LlmSettingsModal
src/components/LlmSummaryMarkdown.tsx	LlmSummaryMarkdownProps, MermaidBlock, LlmSummaryMarkdownInner
src/components/SidebarNav.tsx	SidebarNavProps, SidebarNav
src/vite-env.d.ts	SummaryPrefetchPayload, FileSummaryExportPayload, FileSummaryMarkdownReadPayload, SelectionSummaryExportPayload, SelectionSummaryMarkdownReadPayload, NovadiffDocsWritePayload, NovadiffDocsReadPayload, ElectronAPI

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/prefetch-summaries.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
src/App.tsx	src/app/types.ts
src/App.tsx	src/app/diffExcerpt.ts
src/App.tsx	src/app/llmStorage.ts
src/App.tsx	src/app/selectedDiffDocs.ts
src/App.tsx	src/components/DiffWorkspace.tsx
src/App.tsx	src/components/DocumentationWorkspace.tsx
src/App.tsx	src/components/InsightsColumn.tsx
src/App.tsx	src/components/LlmSettingsModal.tsx
src/App.tsx	src/components/SidebarNav.tsx
src/App.tsx	src/app/novadiffPaths.ts
src/app/changedFilesTree.ts	src/app/types.ts
src/app/codeCityLayout.ts	src/app/types.ts
src/app/codebaseDocs.ts	src/app/types.ts
src/app/commitMessage.ts	src/app/docWorkspaceMetrics.ts
src/app/commitMessage.ts	src/app/types.ts
src/app/diffExcerpt.ts	src/app/types.ts
src/app/docWorkspaceMetrics.ts	src/app/types.ts
src/app/docsQuality.ts	src/app/types.ts
src/app/llmStorage.ts	src/app/types.ts
src/app/novadiffDocs.ts	src/app/types.ts
src/app/selectedDiffDocs.ts	src/app/types.ts
src/components/ChangedFilesTree.tsx	src/app/types.ts
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts
src/components/CodeCityLegend.tsx	src/app/codeCityLayout.ts
src/components/CodeCityView.tsx	src/app/codeCityLayout.ts
src/components/DiffWorkspace.tsx	src/app/types.ts
src/components/DiffWorkspace.tsx	src/components/LlmSummaryMarkdown.tsx
src/components/DocumentationWorkspace.tsx	src/app/types.ts
src/components/DocumentationWorkspace.tsx	src/app/codebaseDocs.ts
src/components/DocumentationWorkspace.tsx	src/app/commitMessage.ts
src/components/DocumentationWorkspace.tsx	src/app/docWorkspaceMetrics.ts
src/components/DocumentationWorkspace.tsx	src/app/novadiffDocs.ts
src/components/DocumentationWorkspace.tsx	src/app/mermaidBoot.ts
src/components/DocumentationWorkspace.tsx	src/app/llmStorage.ts
src/components/DocumentationWorkspace.tsx	src/app/codeCityLayout.ts
src/components/DocumentationWorkspace.tsx	src/app/docsQuality.ts
src/components/DocumentationWorkspace.tsx	src/components/CodeCityLegend.tsx

From	To
src/components/DocumentationWorkspace.tsx	src/components/CodeCityView.tsx
src/components/DocumentationWorkspace.tsx	src/components/LlmSummaryMarkdown.tsx
src/components/InsightsColumn.tsx	src/app/types.ts
src/components/InsightsColumn.tsx	src/app/llmStorage.ts
src/components/InsightsColumn.tsx	src/app/novadiffPaths.ts
src/components/InsightsColumn.tsx	src/components/LlmSummaryMarkdown.tsx
src/components/InsightsColumn.tsx	src/components/ChangedFilesTree.tsx
src/components/LlmSettingsModal.tsx	src/app/llmStorage.ts
src/components/LlmSummaryMarkdown.tsx	src/app/generatedMarkdown.ts
src/components/LlmSummaryMarkdown.tsx	src/app/mermaidBoot.ts
src/components/SidebarNav.tsx	nova-diff-icon.png
src/main.tsx	src/App.tsx
src/vite-env.d.ts	src/app/types.ts
src/vite-env.d.ts	src/app/llmStorage.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/novadiff-docs-html.cjs	src/app/generatedMarkdown.ts	promoteKnownHeadings
electron/novadiff-docs-html.cjs	src/app/generatedMarkdown.ts	splitDenseParagraphs
electron/prefetch-summaries.cjs	electron/compare-runner.cjs	runCompareEngine
electron/prefetch-summaries.cjs	electron/llm.cjs	summarizeChange
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
src/App.tsx	src/app/llmStorage.ts	loadLlmSettings
src/App.tsx	src/app/selectedDiffDocs.ts	buildSelectedDiffDocContext
src/App.tsx	src/app/selectedDiffDocs.ts	isChangedDiffRow
src/App.tsx	src/app/novadiffPaths.ts	isNovadiffDocsReservedPath
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts	buildChangedFilesTreeRoot
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts	flattenVisibleRows
src/components/ChangedFilesTree.tsx	src/app/changedFilesTree.ts	ancestorDirPaths
src/components/DocumentationWorkspace.tsx	electron/file-summary-export.cjs	readFileSummaryMarkdowns
src/components/DocumentationWorkspace.tsx	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
src/components/DocumentationWorkspace.tsx	src/app/docWorkspaceMetrics.ts	buildDocWorkspaceMetrics
src/components/DocumentationWorkspace.tsx	src/app/docWorkspaceMetrics.ts	changeKindPieMermaid
src/components/DocumentationWorkspace.tsx	src/app/docWorkspaceMetrics.ts	depthBarMermaid
src/components/DocumentationWorkspace.tsx	src/app/mermaidBoot.ts	runMermaidNodes
src/components/DocumentationWorkspace.tsx	src/app/codeCityLayout.ts	buildCodeCityLayout
src/components/DocumentationWorkspace.tsx	src/app/commitMessage.ts	buildCommitMessageContext
src/components/DocumentationWorkspace.tsx	src/app/commitMessage.ts	parseCommitMessageOutput
src/components/DocumentationWorkspace.tsx	src/app/docsQuality.ts	deriveConfidenceBadges
src/components/DocumentationWorkspace.tsx	src/app/docsQuality.ts	buildReleaseOverviewMarkdown
src/components/DocumentationWorkspace.tsx	src/app/novadiffDocs.ts	bundleLabel
src/components/DocumentationWorkspace.tsx	src/app/docsQuality.ts	buildSummaryPromptContext
src/components/DocumentationWorkspace.tsx	src/app/docsQuality.ts	buildRiskPromptContext
src/components/DocumentationWorkspace.tsx	src/app/docWorkspaceMetrics.ts	metricsToPromptContext
src/components/DocumentationWorkspace.tsx	src/app/codebaseDocs.ts	buildCodebasePromptContext
src/components/DocumentationWorkspace.tsx	src/app/docWorkspaceMetrics.ts	buildCompareMetricsMarkdown
src/components/DocumentationWorkspace.tsx	src/app/codebaseDocs.ts	buildCodebaseMetricsMarkdown
src/components/DocumentationWorkspace.tsx	src/app/codebaseDocs.ts	codebaseExtensionPieMermaid
src/components/DocumentationWorkspace.tsx	src/app/codebaseDocs.ts	codebaseDepthMermaid

From file	To file	Callee
src/components/DocumentationWorkspace.tsx	src/app/docsQuality.ts	relPathAnchorId
src/components/InsightsColumn.tsx	src/app/novadiffPaths.ts	isNovadiffDocsReservedPath
src/components/LlmSettingsModal.tsx	src/app/llmStorage.ts	saveLlmSettings
src/components/LlmSummaryMarkdown.tsx	src/app/mermaidBoot.ts	runMermaidNodes
src/components/LlmSummaryMarkdown.tsx	src/app/generatedMarkdown.ts	normalizeGeneratedMarkdown

Codebase outline (JSON excerpt)

```
{
  "by_extension": [
    {
      "count": 58,
      "ext": ".pak"
    },
    {
      "count": 20,
      "ext": "(no ext)"
    },
    {
      "count": 18,
      "ext": ".ts"
    },
    {
      "count": 13,
      "ext": ".cjs"
    },
    {
      "count": 11,
      "ext": ".tsx"
    },
    {
      "count": 9,
      "ext": ".plist"
    },
    {
      "count": 5,
      "ext": ".json"
    },
    {
      "count": 4,
      "ext": ".dylib"
    },
    {
      "count": 3,
      "ext": ".svg"
    },
    {
      "count": 2,
      "ext": ".icns"
    },
    {
      "count": 2,
      "ext": ".md"
    },
    {
      "count": 1,
      "ext": ".asar"
    },
    {
      "count": 1,
      "ext": ".bin"
    },
    {
      "count": 1,
      "ext": ".css"
    },
    {
      "count": 1,
      "ext": ".d"
    },
    {
      "count": 1,
      "ext": ".dat"
    },
    {
      "count": 1,
      "ext": ".html"
    },
    {
      "count": 1,
      "ext": ".jpg"
    },
    {
      "count": 1,
```

```

    "ext": ".lock"
  },
  {
    "count": 1,
    "ext": ".nib"
  },
  {
    "count": 1,
    "ext": ".png"
  },
  {
    "count": 1,
    "ext": ".rs"
  },
  {
    "count": 1,
    "ext": ".toml"
  },
  {
    "count": 1,
    "ext": ".yaml"
  }
],
"class_diagram_mermaid": "classDiagram\n class ChangeKind\n class ChangedFilesTree\n class ChangedFilesTreeProps\n class CodeCityFileNode\n class CodeCityFilters\n class CodeCityLayoutResult\n class CodeCityLegend\n class CodeCityModel\n class CodeCityOwnershipSummary\n class CodeCityRenderableBuilding\n class CodeCityRenderableDistrict\n class CodeCitySymbolNode\n class CodeCityView\n class CodebaseOutline\n class DiffRow\n class DiffSelectionLineRange\n class DiffSelectionSymbolMatch\n class DiffSymbolSpan\n class DiffTableRow\n class DiffWorkspace\n class DiffWorkspaceProps\n class DirNode\n class DocWorkspaceMetrics\n class DocumentationWorkspace",
"cross_file_call_edges": [
  {
    "from_file": "electron/code-city-model.cjs",
    "to_file": "electron/git-blame.cjs",
    "via": "blameFileOwnership"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "normalizeMarkdownDocument"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-pdf.cjs",
    "via": "htmlToPdfFile"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/compare-runner.cjs",
    "via": "runCompareEngine"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/llm.cjs",
    "via": "summarizeChange"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/llm.cjs",
    "via": "summarizeChangeStream"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/llm.cjs",
    "via": "probeProvider"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "stopSummaryPrefetchWorker"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "clearSummaryPrefetchCache"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "startSummaryPrefetchWorker"
  },
  {

```

```
"from_file": "electron/main.cjs",
"to_file": "electron/prefetch-summaries.cjs",
"via": "getPrefetchedSummary"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportFileSummaryArtifacts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readFileSummaryMarkdowns"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportSelectionSummaryArtifacts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readSelectionSummaryMarkdowns"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/code-city-model.cjs",
  "via": "buildCodeCityModelPayload"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-writer.cjs",
  "via": "writeNovadiffDocsBundle"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-html.cjs",
  "via": "writeNovadiffDocsHtml"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-pdf.cjs",
  "via": "generateNovadiffDocsPdf"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-html.cjs",
  "via": "refreshNovadiffDocsHtmlFromDisk"
},
{
  "from_file": "electron/novadiff-docs-html.cjs",
  "to_file": "src/app/generatedMarkdown.ts",
  "via": "promoteKnownHeadings"
},
{
  "from_file": "electron/novadiff-docs-html.cjs",
  "to_file": "src/app/generatedMarkdown.ts",
  "via": "splitDenseParagraphs"
},
{
  "from_file": "electron/prefetch-summaries.cjs",
  "to_file": "electron/compare-runner.cjs",
  "via": "runCompareEngine"
},
{
  "from_file": "electron/prefetch-summaries.cjs",
  "to_file": "electron/llm.cjs",
  "via": "summarizeChange"
},
{
  "from_file": "electron/prefetch-summaries.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportFileSummaryArtifacts"
},
{
  "from_file": "src/App.tsx",
  "to_file": "src/app/llmStorage.ts",
  "via": "loadLlmSettings"
},
{
  "from_file": "src/App.tsx",
  "to_file": "src/app/selectedDiffDocs.ts",
```

```

"via": "buildSelectedDiffDocContext"
},
{
  "from_file": "src/App.tsx",
  "to_file": "src/app/selectedDiffDocs.ts",
  "via": "isChangedDiffRow"
},
{
  "from_file": "src/App.tsx",
  "to_file": "src/app/novadiffPaths.ts",
  "via": "isNovadiffDocsReservedPath"
},
{
  "from_file": "src/components/ChangedFilesTree.tsx",
  "to_file": "src/app/changedFilesTree.ts",
  "via": "buildChangedFilesTreeRoot"
},
{
  "from_file": "src/components/ChangedFilesTree.tsx",
  "to_file": "src/app/changedFilesTree.ts",
  "via": "flattenVisibleRows"
},
{
  "from_file": "src/components/ChangedFilesTree.tsx",
  "to_file": "src/app/changedFilesTree.ts",
  "via": "ancestorDirPaths"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readFileSummaryMarkdowns"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readSelectionSummaryMarkdowns"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docWorkspaceMetrics.ts",
  "via": "buildDocWorkspaceMetrics"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docWorkspaceMetrics.ts",
  "via": "changeKindPieMermaid"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docWorkspaceMetrics.ts",
  "via": "depthBarMermaid"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/mermaidBoot.ts",
  "via": "runMermaidNodes"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/codeCityLayout.ts",
  "via": "buildCodeCityLayout"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/commitMessage.ts",
  "via": "buildCommitMessageContext"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/commitMessage.ts",
  "via": "parseCommitMessageOutput"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docsQuality.ts",
  "via": "deriveConfidenceBadges"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docsQuality.ts",
  "via": "buildReleaseOverviewMarkdown"
},

```



```

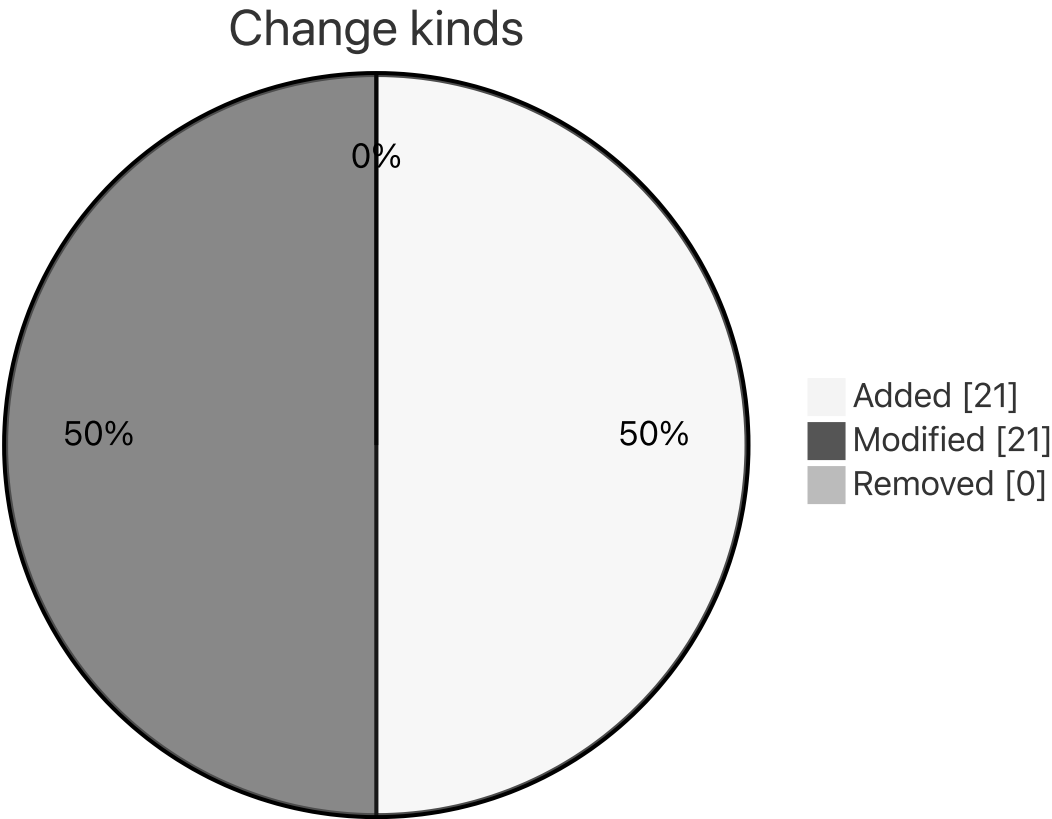
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/novadiffDocs.ts",
  "via": "bundleLabel"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docsQuality.ts",
  "via": "buildSummaryPromptContext"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docsQuality.ts",
  "via": "buildRiskPromptContext"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docWorkspaceMetrics.ts",
  "via": "metricsToPromptContext"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/codebaseDocs.ts",
  "via": "buildCodebasePromptContext"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docWorkspaceMetrics.ts",
  "via": "buildCompareMetricsMarkdown"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/codebaseDocs.ts",
  "via": "buildCodebaseMetricsMarkdown"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/codebaseDocs.ts",
  "via": "codebaseExtensionPieMermaid"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/codebaseDocs.ts",
  "via": "codebaseDepthMermaid"
},
{
  "from_file": "src/components/DocumentationWorkspace.tsx",
  "to_file": "src/app/docsQuality.ts",
  "via": "relPathAnchorId"
},
{
  "from_file": "src/components/InsightsColumn.tsx",
  "to_file": "src/app/novadiffPaths.ts",
  "via": "isNovadiffDocsReservedPath"
},
{
  "from_file": "src/components/LlmSettingsModal.tsx",
  "to_file": "src/app/llmStorage.ts",
  "via": "saveLlmSettings"
},
{
  "from_file": "src/components/LlmSummaryMarkdown.tsx",
  "to_file": "src/app/mermaidBoot.ts",
  "via": "runMermaidNodes"
},
{
  "from_file": "src/components/LlmSummaryMarkdown.tsx",
  "to_file": "src/app/generatedMarkdown.ts",
  "via": "normalizeGeneratedMarkdown"
}
],
"cross_file_call_graph_mermaid": "flowchart LR\n  electron_code_city_model_cjs[\"electron/code-city-model.cjs\"] -->|blameFileOwnership| electron_git_blame_cjs[\"electron/git-blame.cjs\"]\n  electron_file_summary_export_cjs[\"electron/file-summary-export.cjs\"] -->|normalizeMarkdownDocument| electron_novadiff_docs_html_cjs[\"electron/novadiff-docs-html.cjs\"]\n  electron_file_summary_export_cjs[\"electron/file-summary-export.cjs\"] -->|htmlToPdfFile| electron_novadiff_docs_pdf_cjs[\"electron/novadiff-docs-pdf.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|runCompareEngine| electron_compare_runner_cjs[\"electron/compare-runner.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|summarizeChange| electron_llm_cjs[\"electron/llm.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|summarizeChangeStream| electron_llm_cjs[\"electron/llm.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|probeProvider| electron_llm_cjs[\"electron/llm.cjs\"]\n  electron_main_cjs[\"electron/main.cjs\"] -->|stopSummaryPrefetchWorker|

```

```
electron_prefetch_summaries_cjs["electron/prefetch-summaries.cjs"]\n  electron_main_cjs["electron/main.cjs"] --  
>|clearSummaryPrefetchCache| electron_prefetch_summaries_cjs["electron/prefetch-summaries.cjs"]\n  electron_main_cjs["electron/main.cjs"] -->|startSummaryPrefetchWorker|  
electron_prefetch_summaries_cjs["electron/prefetch-summaries.cjs"]\n  electron_main_cjs["e  
... truncated ...
```

Diagrams (Mermaid)

Change mix



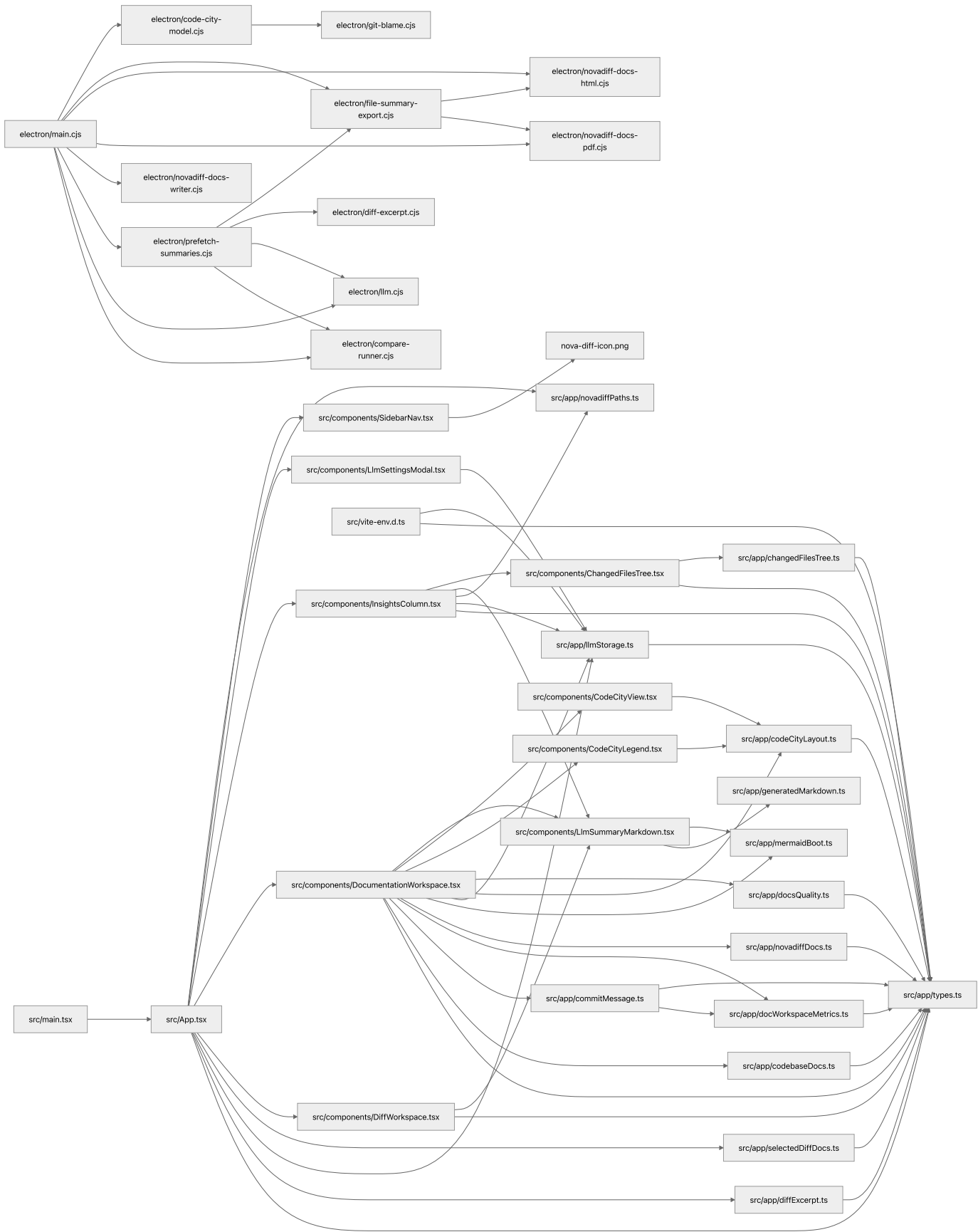
Path depth

depth 1 (3)

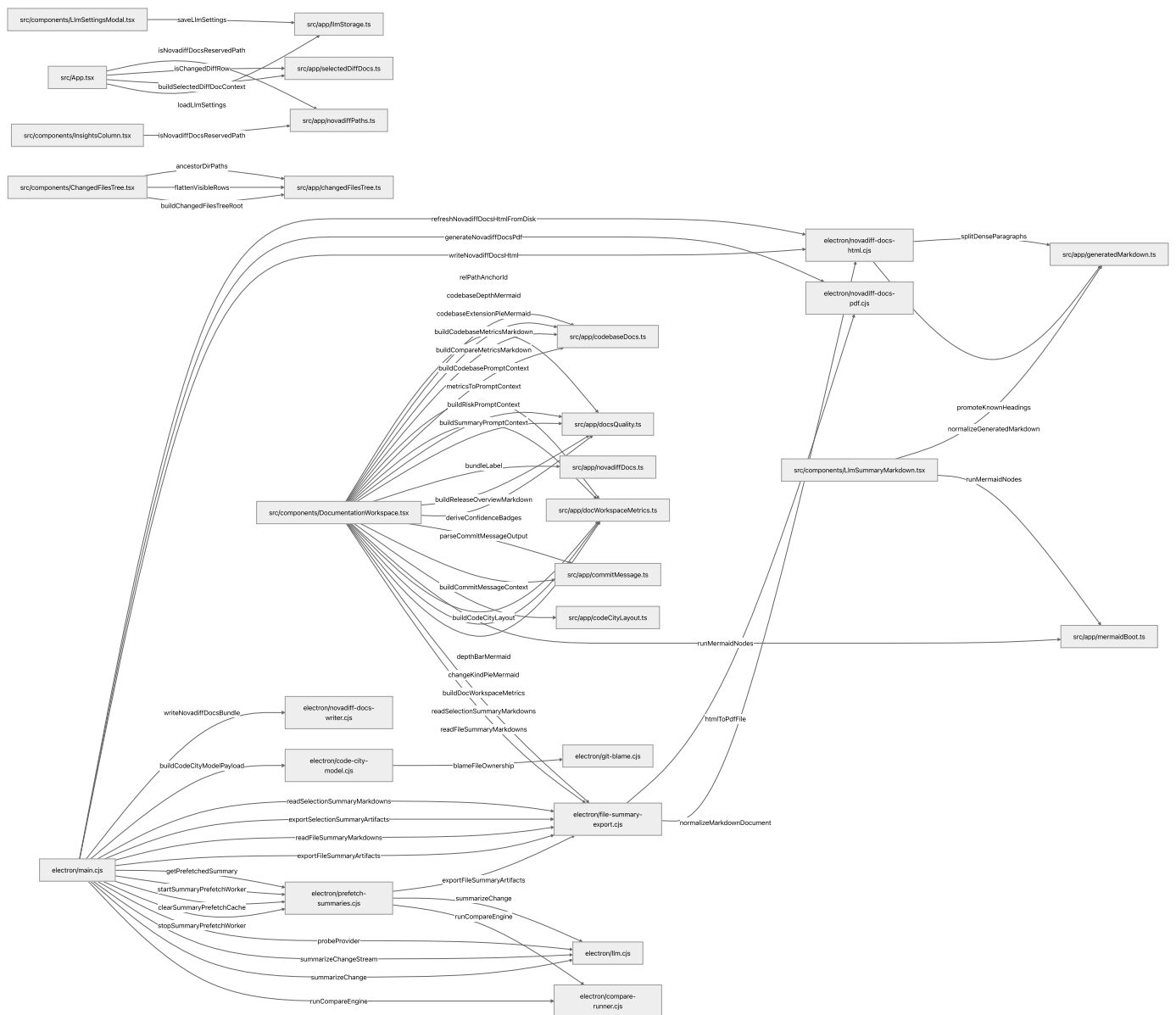
depth 2 (16)

depth 3 (23)

Module import graph (relative imports)



Cross-file calls (heuristic)



Types / classes (regex sketch)

